Pre-Thesis I Report

Detection of DDoS Attacks In Financial Sectors Using Machine Learning

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science

Department of Computer Science and Engineering
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September 2023

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Declaration

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

As the landscape of the digital world continues to evolve, so has the sophistication and complexities of cyber threats and their rate of execution has been boosting precariously. Among all the many diverse industries which are targeted for cyber attacks, the most vulnerable regions include the financial sectors. Cybercriminals attack the banking industry and breach financial data using various strategies, such as DDoS attacks, phishing, malware, intrusion threats, etc. Customers are facing an increase in the rate of cyber crimes, especially DDoS attacks and it is challenging to defend against this crime using conventional non-dynamic algorithms. In this study, we developed a hybrid model by exploring various machine learning and deep learning approaches including neural network-based anomaly detection, CNN and LSTM, feature engineering SVMs, flow-based analysis K-Means, and reinforcement learning to strengthen cybersecurity defense in the banking sector. By leveraging these technologies effectively, financial institutions perform better in bolstering their defenses against an ever-evolving cyber threat scenario, safeguarding their assets, reputation, and the trust of their customers.

Keywords: DDoS Attacks, Data Breach, Detection, Deep Learning, Financial Services, Cybercriminals, Neural Network

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Chapter 1

Introduction

In today's interconnected and digitalized world, the financial sector has become one of the attractive targets for malicious actors seeking to exploit vulnerabilities in cyberspace. The cyberspace term refers to a global digitized sharing environment throughout the world. This includes a vast number of components which play a significant role in data sharing and transfer [5] In this globalized environment most of the major cyber attacks impose threats in the financial sector. The increasing complexity of the financial sector may come up with various cyber attacks like malware, phishing, intrusion, DDoS which also stands out as a huge threat to tackle.

A distributed denial-of-service attack (DDoS) uses a large number of compromised computers (also called botnets) to flood a target's network with a large volume of traffic. It is a different approach to DoS (denial of service) attacks which specifically targets only a single user/network. On the other hand, DDoS is a type of DoS attack that uses multiple networks to generate traffic which is also sent to the target system. DDoS is more complicated than DoS as it's more difficult to defend against the attackers as the traffic comes from multiple sources. [21]
In DDoS the attackers use a huge number of node traffic to compromise their target and later on consume the data from the network. This situation destroys the whole server infrastructure and prevents users from using their network. DDoS attacks have two techniques to implement, one is reflection and other is amplification technique. Using the reflection technique, attackers capture the target's IP and then send packets of traffic by using the amplification technique. There are also 3 types of DDoS attacks, volume based, protocol based and application based. All types have their different target attributes to capture the network system of their target. [21]

The financial sector has been a major target of DDoS attacks over the past few years. They can target any financial institution, including banks, investment firms, and insurance companies. These attacks easily disrupt the critical banking service like online banking, transaction and trading which results in a huge credit loss of the institution. There are several methods to protect against these attacks like robust security, system firewall, intrusion detection and security management. But early anomalies detection can mitigate a vast number of DDoS attackers. In this paper, we proposed a hybrid complex machine learning and deep learning approach to detect the early anomalies in DDoS attacks.

Over the past few years DDoS attacks have become a common threat in IT security. According to some reports, DDoS growth lasted over 50 hours in 2022 compared with 2021 where they lasted only for 30 minutes.The increasing traffic jamming in the network causes a numerous number of losses. According to InfoSecurity Magazine Report, DDoS attacks rose around 31% in Q1 of 2021 over the same period of 2020 for 2.9 million of attacks. [9] Moreover, DDoS has become a common threat to different business sharing and e-commerce companies like Amazon and EXMO Cryptocurrency Exchange. In February, 2020, Amazon Web Service (AWS) faced the biggest DDoS attack consisting of a 2.3Tbps of network which jammed their website and affected an immense number of buyers. [23].

On the other hand, in February of 2021, DDoS took down one of the greatest British cryptocurrency exchanges EXMO by shutting their website down temporarily for two hours. The company reported a huge traffic problem where a numerous number of connections disrupted their service. They also stated that the traffic was so massive that drove around 30 GB connections per second which affected the server's whole network system. Later the attack was tackled by a DDoS protection expert. EXMO assumed that the attack took place because of revealing any private keys. [24] [12]

One of the reports from PRNewswire shows that 22% of DDoS attacks target any financial sector. It's spreading through Europe where almost 73% of cyber attacks have increased and half of the DDoS attacks are targeted to the financial companies. The report also derives that, most of the companies now focusing on extortion tactics. [26]

In 2022, the DDoS attacks significantly increased in the UK financial sector which were mostly driven through the Russia-Ukraine war. One of the reports by Scroxton stated that the data of financial states may be targeted by the hackvisits of the ongoing war. He also mentioned that the ransomware attackers are creating additional tactics to implement DDoS attacks into their plan. In their tactics, multiple IP has been compromised using very fast but weighted traffic signals which makes the protection more difficult to tackle. [24]

To tackle DDoS attacks, an individual must detect the attack as soon as possible. DDoS detection is the process that distinguishes DDoS attacks from normal cyber attacks. There are many methods that are applied today which are becoming more and more sophisticated. To detect the attack, an individual must gain productive knowledge about the whole networking system. Then different types of machine learning and deep learning needed to be applied. Different models like CNN, LSTM or ANFIS can be applied to detect DDoS among more cyber attacks. In order to classify the data, the SVM model is very popular as it gives very accurate results. In summary, hybrid machine and deep learning models can be easily implemented to detect any DDoS attack very precisely.

1.1 Research Problem

Just as the digitalization of technology keeps carrying forward, the rate of threats towards technological firms and financial services also tend to keep boosting. Previously, denial of service(DoS) attacks played a malicious role in the world of cyber crime but gradually, DoS also began to enlarge and develop in large scales. According to [27], denial of service is an attack which aims to shut down a machine or network, making it inaccessible to the intended user. Victims of DoS attacks often target web servers of high-profile organizations such as banks, commercial and media companies, or government and commercial organizations. Although DoS attacks do not typically result in the theft or loss of critical information or other assets, managing them can cost victims significant time and money. On the contrary, the improved version of DoS, popularly known as distributed denial of service(DDoS), tends to attack financial services in a similar procedure. [25] In addition, DDoS uses multiple computers or machines for flooding a targeted resource. Both types of attacks overload a server or web application in an attempt to disrupt services. [3] The prime objective of this commonly used cyber threat is to reduce the availability of services which results in losses of money and a variety of other issues. In [6] Norton broke down the categories of DDoS attacks into two different types based on the intended outcomes of the hackers.

- **Flooding attack:** This refers to an attempt of applying overwhelming flood of data to take down a server. An ICMP or ping flood occurs when an attacker or hacker sends data packets to affect an entire group of connected computers, thereby bringing down that misconfigured network.
- **Crashing attack:** The rare DDoS variant involves blocking a compromised system with bugs that affect weaknesses in the targeted infrastructure. Once these unpatched vulnerabilities are exploited, the system will crash.

The preliminary methods for detecting and preventing DDoS attacks are traffic analysis, anomaly detection, rate limiting, signature-based detection, behavioral analysis, traffic filtering, flow analysis, CAPTCHA, traffic shaping, CDN, anycast routing, and many more. However, there are many problems with this detection analysis method. At first, this can make the detection slow. The traditional detection methods mostly rely on known statistical analysis or signature patterns. A new type of DDoS attack might not be known making them difficult to detect in the traditional way, they can generate a high number of false positives and can be difficult to scale large networks [27]. Machine learning and Deep learning algorithms are used to detect this problem in a more efficient approach [2]. Large datasets of historical network traffic, including both malicious and lawful traffic, can be used to train hybrid models using machine learning and deep learning algorithms. This enables them to learn the characteristics of harmful traffic and accurately discriminate between malicious and regular traffic. Those models can improve accuracy, reduce false positives, and have the ability of scalability. ML and DL algorithms for instance Decision tree (DT), Random forest (RF), ANN, KNN, SVM, Adaptive Boosting (AdaBoost), Autoencoder (AE), Convolutional Neural Network (CNN),

long short-term memory networks(LSTM), and Support Vector Machine (SVM) [19].

CNN can be used to monitor network patterns of traffic which are the signs of DDoS attacks and it can be trained to detect patterns in network traffic packet source IP addresses that indicate a SYN flood attack [20]. Since CNN models are not good with non-image data, the data should be encoded and converted into an image [17]. This is going to ease the task of CNN to classify the DDoS attacks resulting in building mitigation systems. Furthermore, LSTM can be recognized by the received number of packets per second which is an unusual traffic pattern [22]. According to [18], LSTM, developed by Hochreiter and Schmidhuber, is a unique RNN architecture which is able to learn long-term dependencies. However, the ability of LSTM depends on how the parameters have been chosen. Lastly, SVM is a classifier that can detect network traffic as legitimate depending on the source IP addresses, destination IP addresses, and port numbers [10]. In [14], it is observed that when compared to other models such as KNN and RF, SVM is a more robust technique. But the benefit of using SVM is its efficiency and accuracy in classifying DDoS attacks. In conclusion, for a more robust and comprehensive detection system, we can develop a hybrid model that contains CNN, LSTM, and SVM. This integrated technique allows for the evaluation of DDoS samples from numerous angles, resulting in increased accuracy, early detection, and a better understanding of attack behavior.

1.2 Research Objectives

The aim of our research is to develop a DDoS attack detection system particularly for the banking industry using machine learning approaches. To build the system, we created a hybrid model merging ML and DL algorithms, specifically we applied CNN, LSTM and SVM. It is very crucial for a bank to have continuous service and a secure system. On the other hand, banking services are seriously threatened by DDoS attacks. The objective of this research are:

1. To get an in-depth knowledge of the specific challenges and complications of distributed denial of service (DDoS) attacks in the banking sector, including methods of attack and previous attack data.

2. To analyze and comprehensively understand several machine learning techniques that are appropriate for DDoS detection, taking into account aspects like real-time processing and scalability to changing attack tactics.

3. To create a hybrid model using CNN, LSTM and SVM for the banking industry that uses pertinent information taken from network traffic data to detect DDoS attacks before they do significant damage.

4. To carefully evaluate the model's performance in the actual world, taking into account variables like accuracy, precision, recall, and F1 score.

5. To offer improvements for enhancing the model.

Chapter 2

Literature Review

Islam et al. [14](2022) proposed a hybrid Machine Learning method to detect distributed denial-of-service (DDoS) on banks and other financial organizations. In the paper, they used various classification models for detecting DDoS attacks. They applied various models like SVM, KNN and RF which all have accuracy more than 97.5% and concluded that SVM has more accuracy than the other two. In this paper, they have a banking dataset where they pre-processed the dataset and then scale to balance the features. Using the feature selection, they applied a machine learning model by splitting the data by 70% of the train set and 30% of the test set. Implementing SVM into the model by a general novel architecture they find out the hyperline between normal and DDoS attacked data. Random forest models are used to detect DDoS attacks using RF classifiers. Also KNN and perfection matrix implemented in the chosen dataset.The main issue with this paper is the whole paper mostly focuses on the accuracy of trained models. But the accuracy can be varied and can be highly achieved by other precision and f-score. They further studied the performance of the models and found that SVM was the best performing model in terms of accuracy. They also concluded that the models can be used effectively to detect DDoS attacks.

Mehmood et al. [11](2021) introduced a hybrid model like Hidden Markov Model (HMM) which is used for network intrusion detection (NIDS) in banking for fraud prevention. The approach comprises three different hybrid approaches for detection and categorization. They use SVM for detecting any intrusion and then use ANFIS to categorize different types of intrusion attacks. The ANFIS also detects DDoS attacks among some various malicious attacks like PROBE, R2U and U2R. In this paper, the authors later distinguish the attacks by FGSVM classifier which gives the best accuracy for tuning the dataset which gives the result of accuracy, recall, precision and F-score by using a proposed formula. In this paper, using the NSL-KDD dataset they implemented 20% training dataset along with 21% test data and 15% data validation. Using this hybrid model, the bank system implements an OTP for every customer who is registered into their system and ensures that only legal types of transactions are being passed through their system. The main issue that arises in this paper is the complexity of using different hybrid models in only a single system. Critics may argue that such complexity could make it challenging to implement and maintain the system. A DDoS attack is an example of an area where the HMM model is accurate and has been used to tackle cyber attacks.

Alghazzawi et al. [7](2021) emphasized a hybrid improved model to detect DDoS attack by using machine and deep learning because of inherent exceptions of these models. They implemented hybrid deep learning models like CNN and BiLSTM for detecting DDoS most accurately and effectively. The hybrid CNN-BI-LSTM model has an accuracy of 94.52%. The authors focused mostly on a benchmarked dataset which later was pre-processed and selected the features by concentrating on the DDoS attacks. The data in the study were embedded using Keras Embedding Layer then dropout based on the BOR strategy. Each data was passed through different layers for example, convolutional layer, pooling layer, extraction layer. Then using the SoftMax function, the data determined the probability of DDoS attacks. The result shows a 21% value with a normal traffic system. The experiment was succeeded by different techniques like confusion matrix, accuracy, precision and f-score. The best accuracy was so far shown by the SVM which has 74% of accuracy in this data model. This hybrid model has a very good successful rate by detecting DDoS attacks. The author also stated some limitations like, using a single dataset like CICDDoS2019, using a single statistical figure, lack of pre-trained CNN model and using binary classification technique for categorizing selection.

Kathirkamanathan et al. [15](2022) initiated three deep neural networks in order to detect distributed denial-of-service (DDoS) on financial services. A high accuracy deep-learning model has been generated in this research to mitigate the existence of DDoS attacks which take place in the application layer of the endpoints of a transaction. In order to construct the model, Keras and TensorFlow had been applied. There are two vital techniques for detecting the existence of distributed denial-of-service (DDoS) attacks in banking and other financial organizations. One is the signature based attack detection method which is functional especially upon well-known attacks. On the other hand, an anomaly-based method is applied for unknown types of threats of DDoS. At first, using past DDoS attacks, the machine learning model was developed. Among the machine learning algorithms LSTM had been considered as the most efficient one. A stack of LSTM was developed for creating a deep-learning model containing multiple layers with various functions. It was observed that using the Stacked LSTM model, more accurate and precise results were achieved (99.3% of accuracy). The outcomes were received using Keras and TensorFlow. Thus, they concluded that this deep learning model is the most efficient option for detecting DDoS attacks. But as the model becomes more complex, there is a higher risk of over-lifting , especially if the dataset is limited. Furthermore, longer training times can delay the deployment of the model causing the DDoS attacks to evolve frequently. However, this model can be automated in the future in order to improve performance parameters such as accuracy, precision, etc.

Wei et al. [13](2021) proposed to combine two deep learning-based models and produce a hybrid model in order to effectively detect distributed denial-of-service(DDoS) attacks and perform classification. The hybrid model is named as AE-MLP which is created by the combination of an Autoencoder(AE) and a Multi-layer Perceptron (MLP). The hybrid model is not only used for detecting the attacks but also to classify the attack into distinct attack types in a timely manner. The proposed model uses an autoencoder (AE) to extract the most significant features from large-scale

DDoS attack datasets. Finding the most suitable compressed and minimized feature set for detecting malicious payloads not only improves detection accuracy but can also effectively reduce costly execution time. Since the model can also classify attacks into various types, this classification capability can provide cybersecurity professionals with the opportunity to design an optimal and appropriate response strategy as quickly as possible before disastrous damage is caused by different types of DDoS attacks. When the outcomes were extracted, it was observed that the accuracy rate was 98%. Despite the high accuracy rate of the approach, combining different architectures may require additional efforts in terms of hyperparameter tuning and model integration. Moreover, AE and MLP components may not always work correctly together or have compatibility issues. But although this model might elevate the rate of latency, this model is still useful in terms of tracking DDoS attacks and also distinguishing them into different categories.

Aysa et al. [1](2020) proposed a framework for the detection and abnormal activities on IoT devices. They used many machine learning and data mining techniques. But among them, they found that mixed random forest (RF) and decision trees (DT) achieved high accuracy in detecting DDoS attacks. For the working phase, they divided the framework into four phases. They used algorithms for data cleaning,data-normalization and feature selections. From the result, they found that True Positive (TP) and True Negative (TN) predictions, which indicate accurate detection and non-detection, are regarded as desirable. False Positives and False Negatives (FP and FN) are both undesirable. The accuracy and recall (sensitivity) metrics, as well as the F-score and Mean Absolute Error (MAE) as performance measurements, are discussed. When different classifiers were applied to different datasets, Decision Tree (J-48) and Random Forest produced higher True Positive Rates (TPR) and faster processing times than the Neural BP network and LSVM. For the dataset, they collected data from three IoT devices. A framework for anomalous activity detection was developed in the field of IoT device security, showing strengths such as algorithm efficacy, organized stages, and thorough metric utilization. However, the framework's adaptability and robustness are called into question due to ambiguity in method explanation, potential dataset specificity, and a lack of validation techniques.

Sambangi et al. [4](2020) proposed a machine learning approach to detect DDoS cyber attacks on financial institutions by applying Multiple Linear Regression (MLR) Model. They divided the detection into two steps: IG Entropy and Traffic Collection Module. The authors used the CCIDS 2017 dataset for preprocessing. Firstly, they used the Traffic Collection Module in order to extract those messages and entries of a device that are intrigued with DDoS threats. This extraction is to recognize the DDoS attack. Secondly, they implemented IG Entropy in order to select the features of the CCIDS 2017 dataset and derive the important attributes that are better deliverables for the prediction model. Then they used the derived attributes for performing a MLR to examine the behavior of selected attributes within the CICIDS 2017 dataset. They performed the MLR model on the significant attributes by analyzing the fit charts and residual plots. After reviewing the paper we see that their proposed model isn't a hybrid model for which they had easily trained and tested their data. As a result, the accuracy at the first test gave an accuracy

of 97.86%. The shortcoming of the paper is when the proposed model is trained after a some moment the accuracy level declines to 73.79%. Moreover, their model is much simpler for detecting such DDoS type attacks as the DDoS attack too has many complex subtypes.

Gill et al. [21](2023) uses a variety of classification algorithms to forecast and categorize financial oriented cyber attacks in order to address the serious threat posed by cyberattacks, particularly on banks due to their valuable data.They perform support vector machine (SVM) in order to increase model complexity which gives detection accuracy of 99.5%. According to their research, SVM is the most trustworthy method compared to KNN,RF, for identifying cyberattacks in the financial industry. First of all, they collected a fraud-detection dataset which contains nasty virus DoS. Then they preprocessed the data and other necessary steps like correlation matrix to make the dataset more valid. After that they split the dataset and applied machine data training models using SVM, Random Forest, KNN in order to measure the performance. They also figured out the time complexity of the model and concluded that SVM is the most effective approach. However, these systems can only look at historical data, so there is still an issue. Therefore, we require something that can defend against attacks which are real-time taking place.

Razib et al. [16](2022) address the problems caused by the Internet of Things (IoT) explosive growth and the requirement for reliable security solutions. The researchers suggest using an intrusion detection system (IDS) driven by Deep Learning (DL) and Software Defined Networking (SDN), named DNS LSTM to counter cyber risks in the IoT. They train the model on the CICIDS 2018 dataset and assess its performance using important criteria like Accuracy, Precision, Recall, and F1-Score. They also evaluates RNN LSTM against other classifiers including DNNGRU and BLSTM to confirm its efficacy, Astonishingly, the suggested system exceeds currently available solutions in the literature, attaining outstanding results like 99.55% accuracy, 99.36% precision, 99.44% recall, and 99.42% F1-score. They show that RNN LSTM is an effective solution for dealing with cybersecurity issues in the IoT environment. Although the approach utilizing RNN LSTM yields impressive results, some potential weaknesses include the limited evaluation scope due to the heavy reliance on a single dataset (CICIDS 2018), which might not cover all potential IoT attack scenarios, and the lack of consideration for real-world deployment challenges and scalability outside of the research environment.

Chen et al. [8](2021) proposed Autoencoder and GRU model for deep learning approach to detect DDoS attack. They used the data set CIC-IDS 2017 for preprocessing. The autoencoder technique is used to extract the features and recreating a data packet of low and uniform length. After preprocessing, they used a two-layered stacked GRU model for training and as a result they extracted time sequence characteristics as GRU networks are good at capturing dependencies in sequential data. Finally by differentiating the trained and raw characteristics, the DDoS attack is detected. The proposed technique is a time saving technique as the approach doesn't depend on manual feature engineering. Moreover the GRU model continuously gives more than 99% accuracy. However, this model is a bit complex as it calculates time sequence using many complex gradient functions.

Chapter 3

Work Plan

To encounter the real world information based network data that are infected with DDoS, we propose a precise and vigorous hybrid model that encompasses the machine learning approach with its subset which is deep learning approach. Our hybrid model comprises one machine learning technique (SVM) and two deep learning techniques (CNN and LSTM). To implement our hybrid model, we used a dataset resembling the true real-world data. We first collected real world data and selected some labeled features namely time stamp, source, and destination IPs, source and destination ports, protocols and types of attack. We then encoded the selected features using autoencoder and then those features were preprocessed. This preprocessed dataset is then splitted into training and testing data sets in order to implement the mentioned techniques. The training dataset is used to train our hybrid model and the testing dataset is used to determine the accuracy and precision.

We develop a Convolution Neural Network (CNN) considering several key steps including data preprocessing, model architecture, training, and evaluation. The convolutional layer in CNN performs for feature extraction. We create our architecture of CNN model including several components like convolutional layers, activation functions, pooling layers, fully connected layers. Our CNN model was trained using Keras library with TensorFlow backend. We used it for DDoS attack classification. Lastly, we evaluate our mode by calculating various metrics like accuracy, precision, recall and F-1 score.

Secondly, CNN extracts features from network traffic data. Then the output of the CNN fed into the LSTM.It learns from network traffic data long-term dependencies. Output of the LSTM is routed to a fully connected layer, which categorizes network traffic as malicious or non-malicious.It is a powerful tool for detecting attacks effectively in conjunction with CNN.

Finally, to evaluate the performance and efficiency level of our proposed hybrid model, we executed the SVM algorithm and carried out the regression analysis. We trained the SVM model using the training dataset. Some common kernel functions are used for SVM and when our training data is trained using SVM model. During the training, SVM determines a function that fits the training data. That function is tuned to optimize our model's performance.

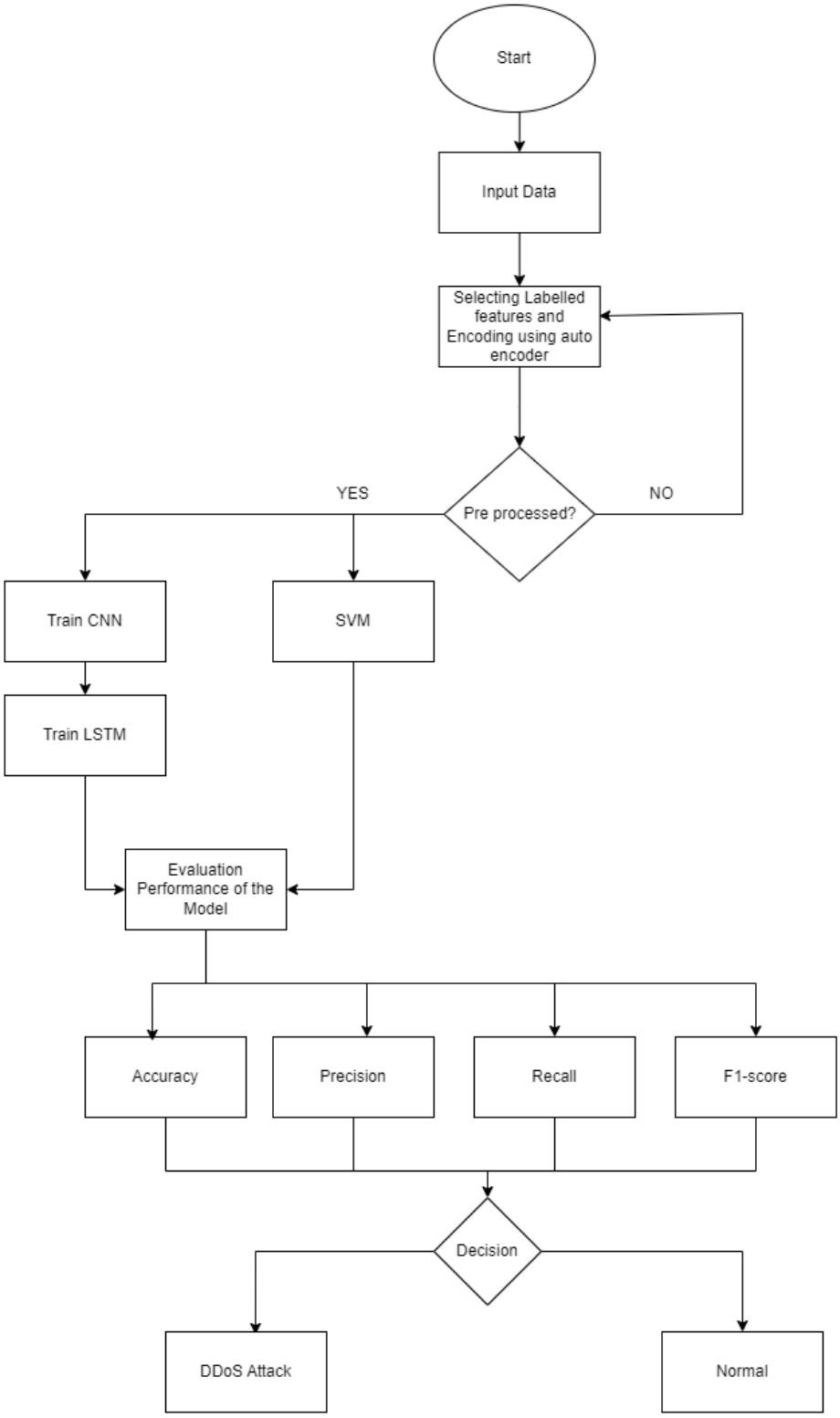


Fig 3.1: Work Flow

Chapter 4

Conclusion

Due to quick growth of various cyber threats and the growing skill of cyber attackers, the field of cyber threat detection in financial sectors has made tremendous advancements. DDoS attacks are a popular cyber threat in financial sectors because their way of breaching data is very much diversified. Besides, the DDoS attacks frequently target the financial services sector as their target. Moreover, the cybercriminals who develop DDoS implement various types of tactics in order to avoid being detected by security agencies and software. To counter these tactics and unprecedented challenges, our research has developed a hybrid model by integrating ML and DL approaches (integrating CNN, LSTM and SVM). We used CNN for specifying the structural patterns and correlations in network traffic, LSTM for designing the sequential data model to analyze the incoming network data in real time and finally SVM for regression analysis like calculating accuracy, precision score through training the labeled historical network data. In essence, our goal of integration of the mentioned techniques is to maximize the chance of DDoS detection in financial sectors is maximized without detecting it by some unknown external software and programs.

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